RAK15004 WisBlock FRAM Module Datasheet

Overview

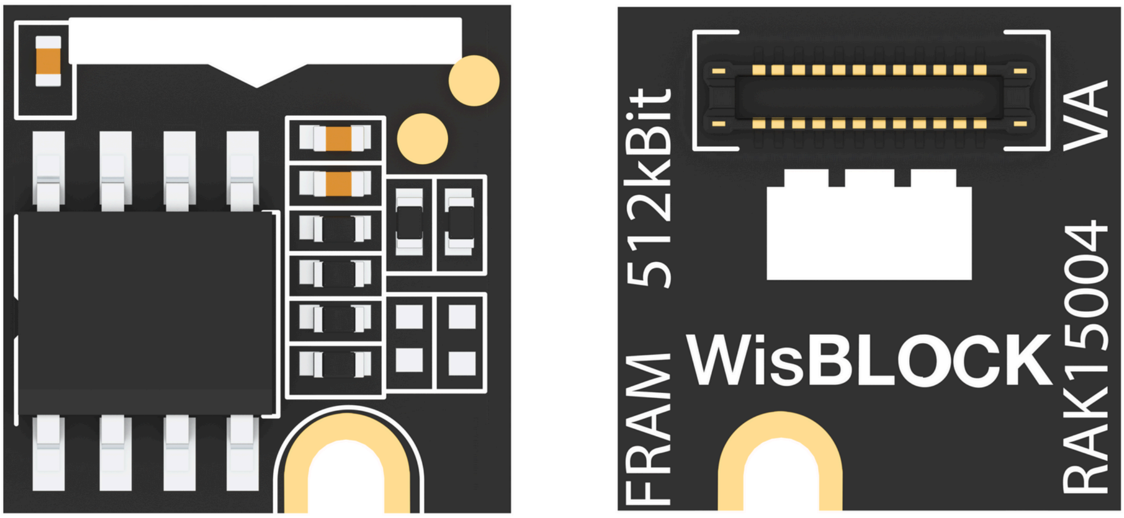

Figure 1: RAK15004 WisBlock FRAM Module

Description

RAK15004 is a WisBlock FRAM Storage Module based on MB85RC512T 512 kbit (64k x 8) from FUJITSU. It is a very high endurance nonvolatile memory storage chip that provides a write/read count of 10,000,000,000,000 per byte. It can be interfaced via I2C and support High-Speed Mode at 3.4 MHz.

Features

- Sensor specifications
  - Chipset: **FUJITSU MB85RC512T**
  - Voltage supply: **3.3 V**
  - Operating current: **15 µA ~ 0.71 mA**

  - 65536 words × 8 bits
  - High Reliability:
    - Read/write endurance: **10,000,000,000,000 / byte**
    - Data retention: **10 years (+ 85° C) & 95 years (+ 55° C)**
  - I2C digital interface and supports High Speed Mode at **3.4 Mhz**
  - Operating temperature: **-40° C ~ 85° C**

- **Module size**

  - 10 × 10 mm

Specifications

Overview

Mounting

Figure 1 shows the mounting mechanism of the RAK15004 module on a WisBlock Base board. The RAK15004 module can be mounted on slots: **A, C, D, E, and F**.

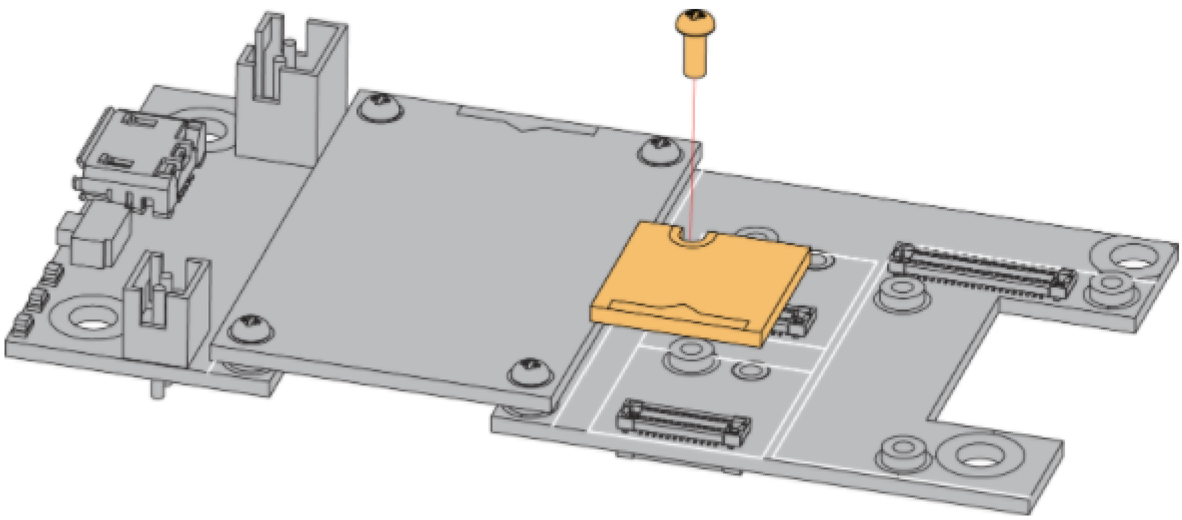

Figure 2: RAK15004 WisBlock FRAM module mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, and their corresponding functions and diagrams. It also covers the electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK15004 WisBlock FRAM Module.

Chipset

Vendor	Part Number
FUJITSU	MB85RC512T

Figure 2 shows MB85RC512T the device addressing, which A1 and A2 were pulled up to high level. The device address is 1010110 (0x56).

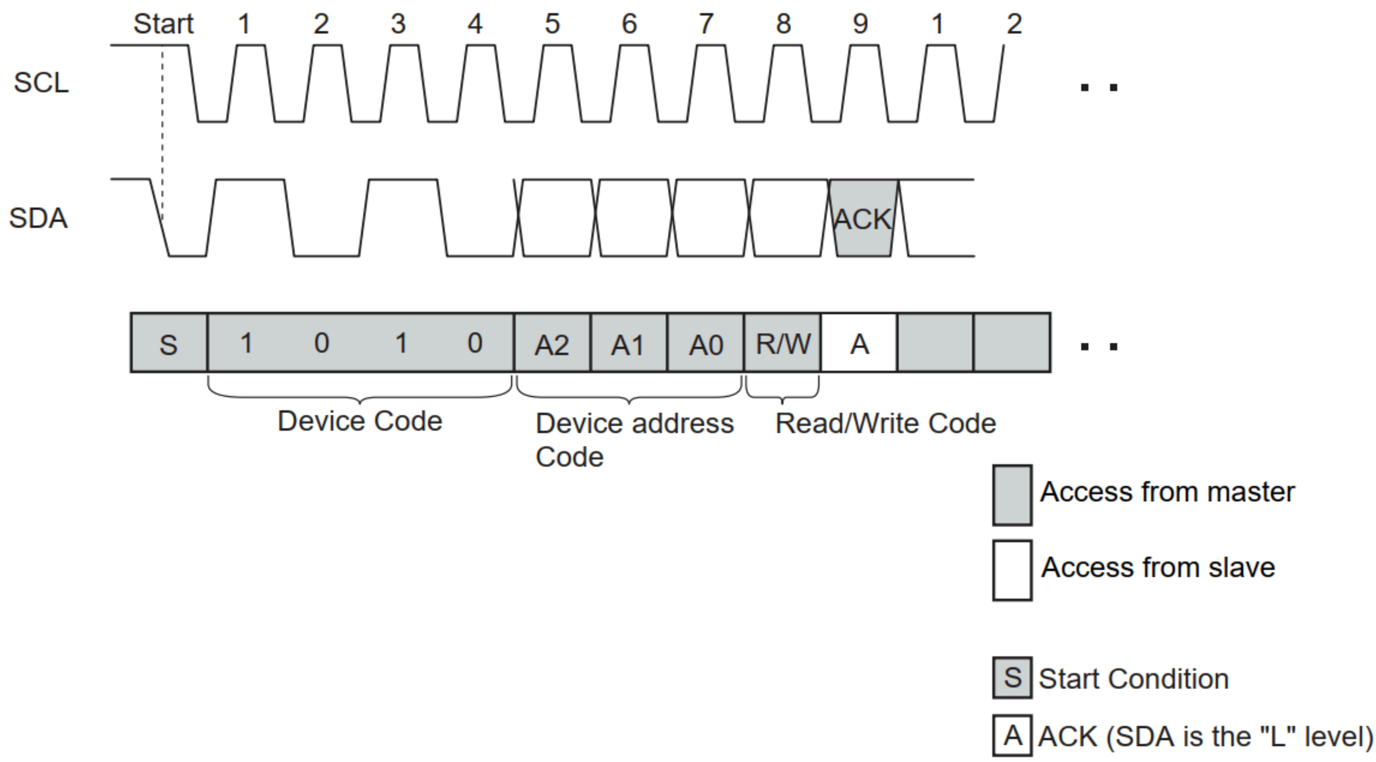

Figure 3: MB85RC512T device addressing

Pin Definition

The RAK15004 WisBlock FRAM Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK15004 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 3**.

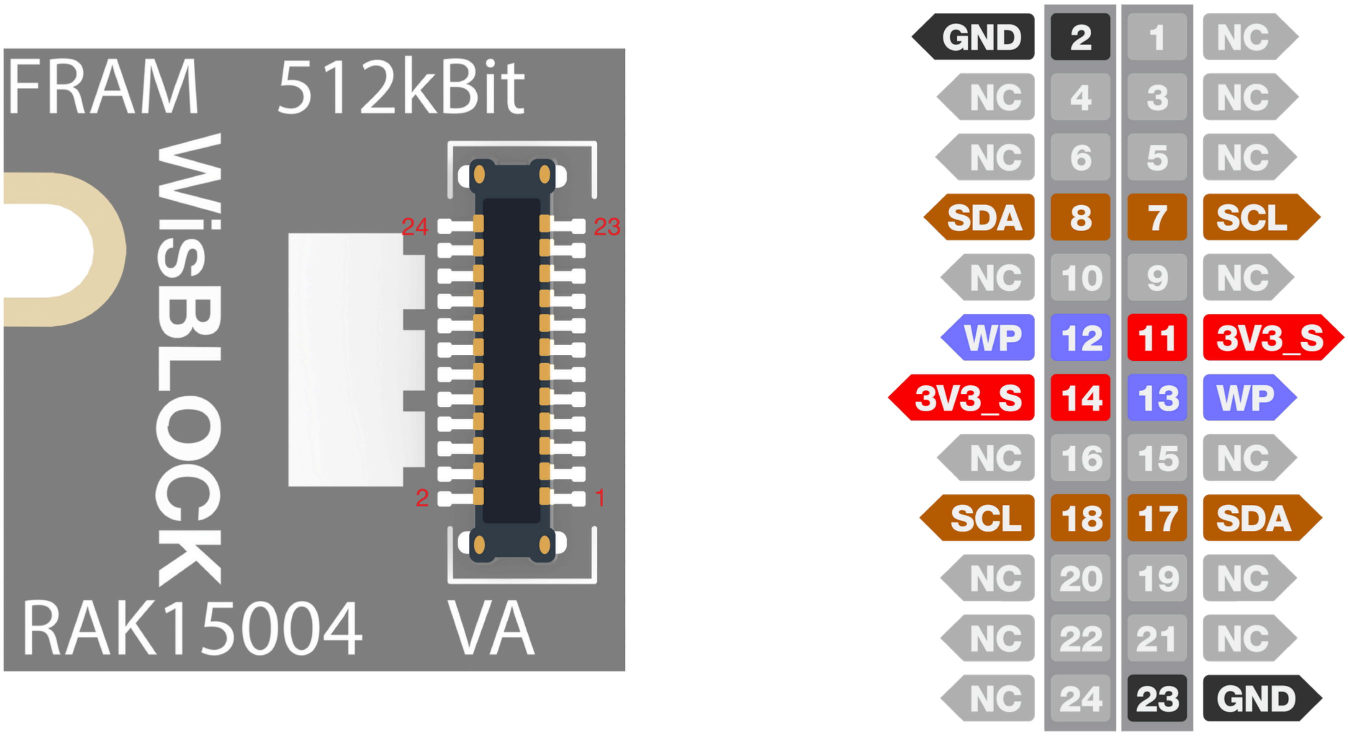

Figure 4: RAK15004 pinout diagram

> **NOTE**
>
> - Only **I2C** related pin, **3V3_S**, and **GND** are connected to WisBlock connector.
> - **3V3_S** voltage output from the WisBlock Base that powers the RAK15004 module can be controlled by the WisBlock Core via `WB_IO2 (WisBlock IO2 pin)`. This makes the module ideal for low-power IoT projects since the WisBlock Core can be disconnected from the RAK15004 module power.

Electrical Characteristics

Symbol	Description	Min.	Nom.	Max.	Unit
VDD	Power Supply Voltage	-	3.3	-	V
I_{stb}	Standby Current	-	-	15	uA

Symbol	Description	Min.	Nom.	Max.	Unit
I_{CC}	Supply Current (Read & Write)	-	-	710	uA

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK15004 module.

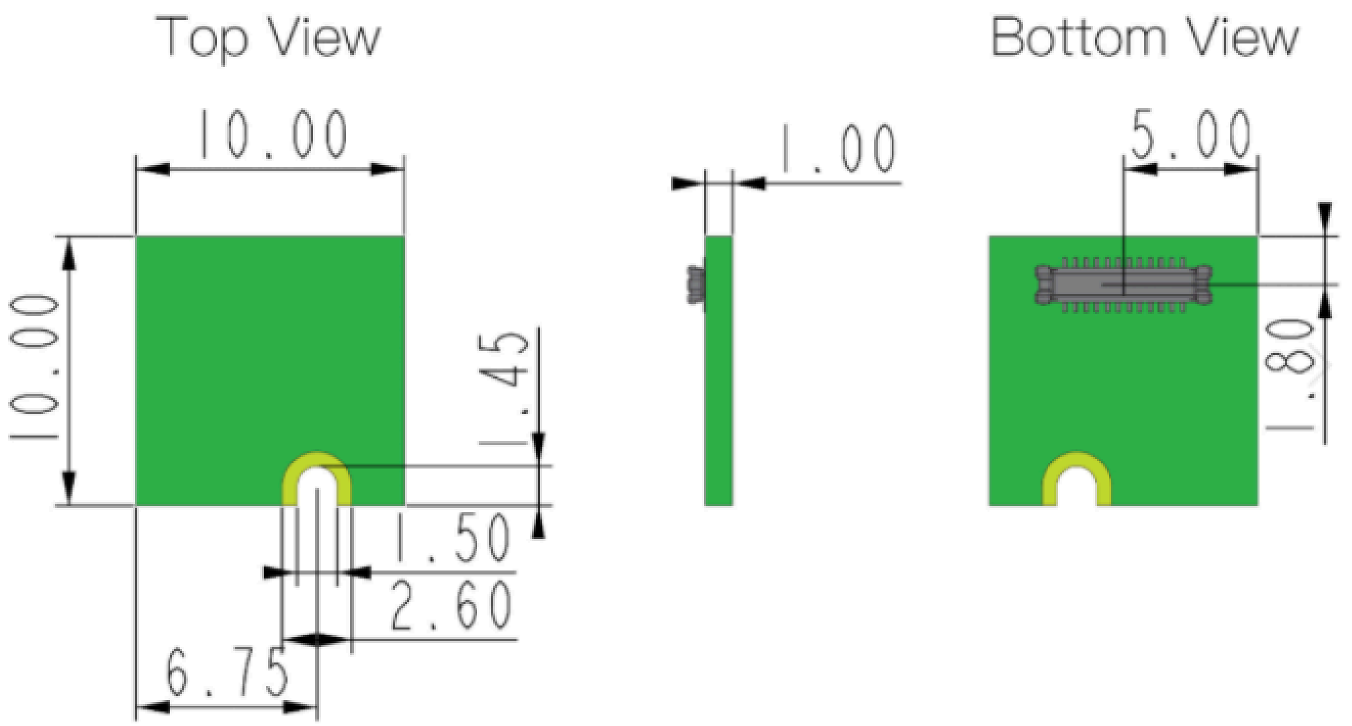

Figure 5: RAK15004 mechanical dimensions

WisBlock Connector PCB Layout

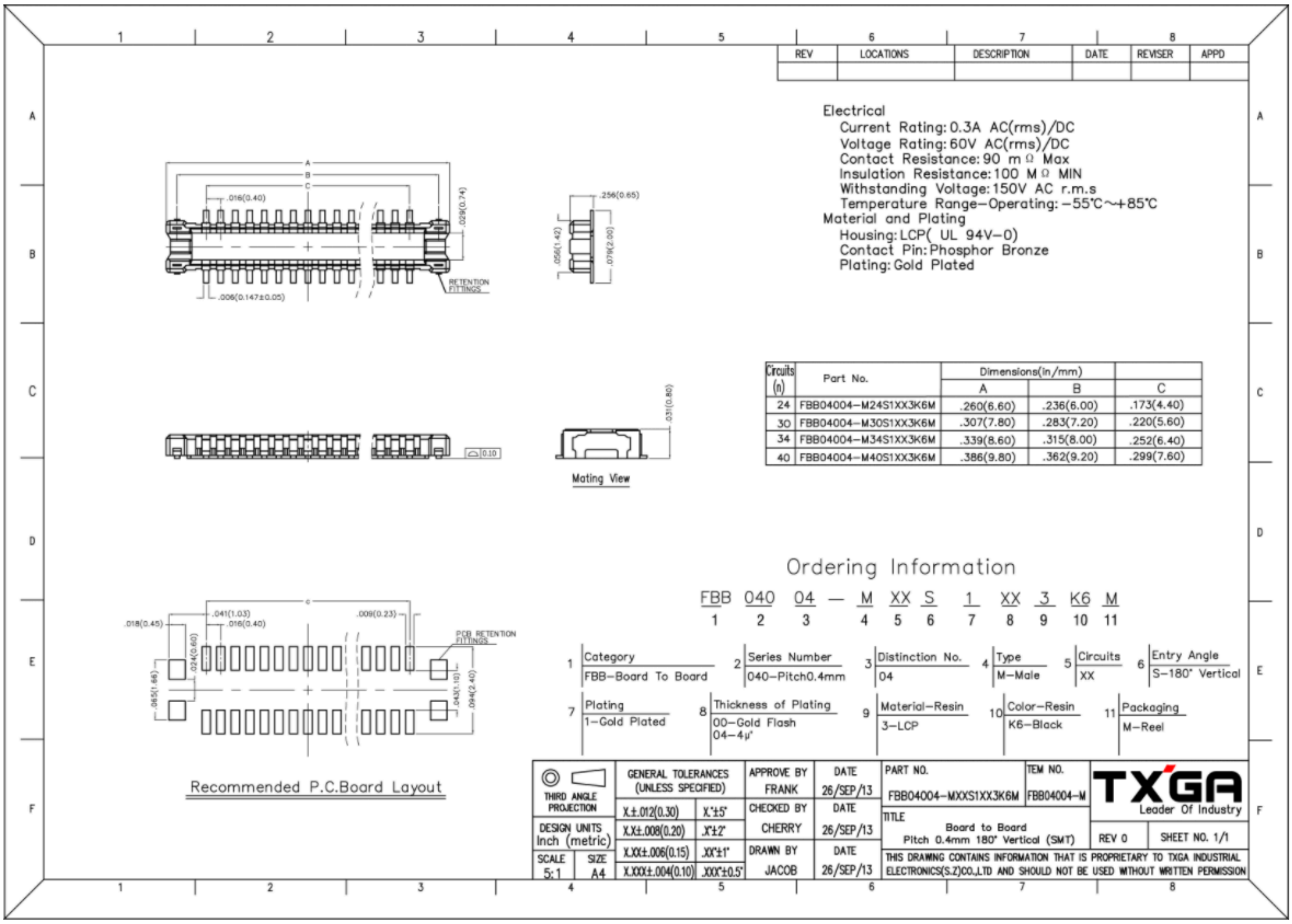

Figure 6: WisBlock Connector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK15004 FRAM module. The default I2C address of the FRAM module is **0x56**.

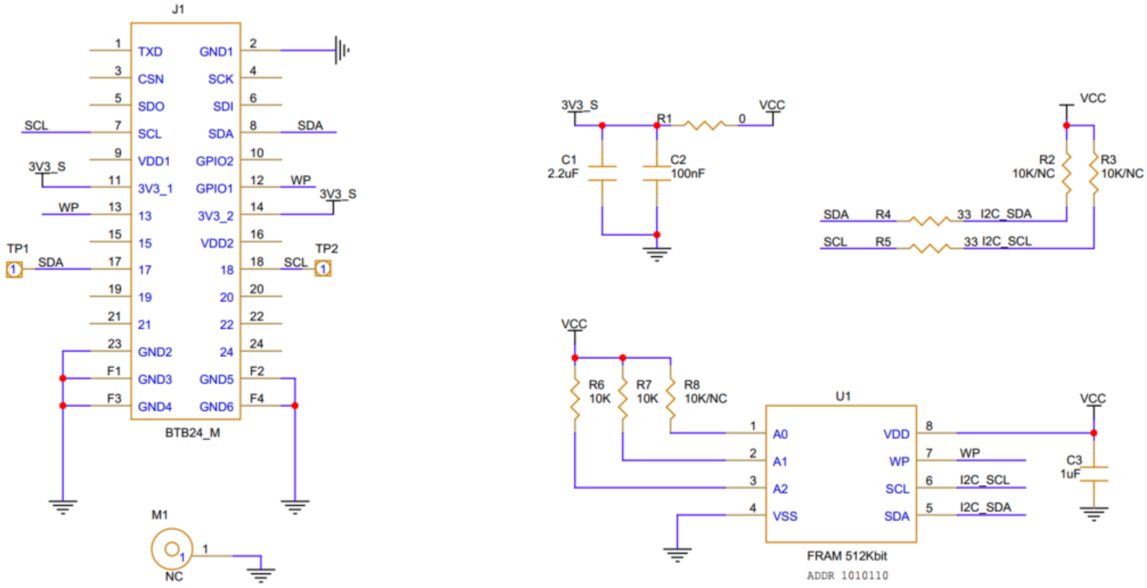

Figure 7: RAK15004 schematic diagram